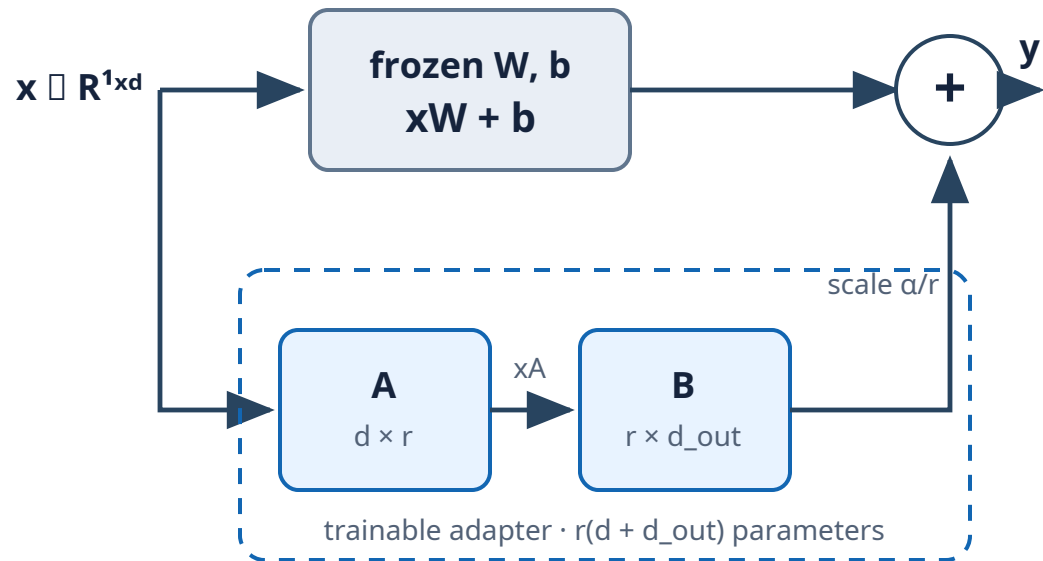


LoRA

linear low-rank correction

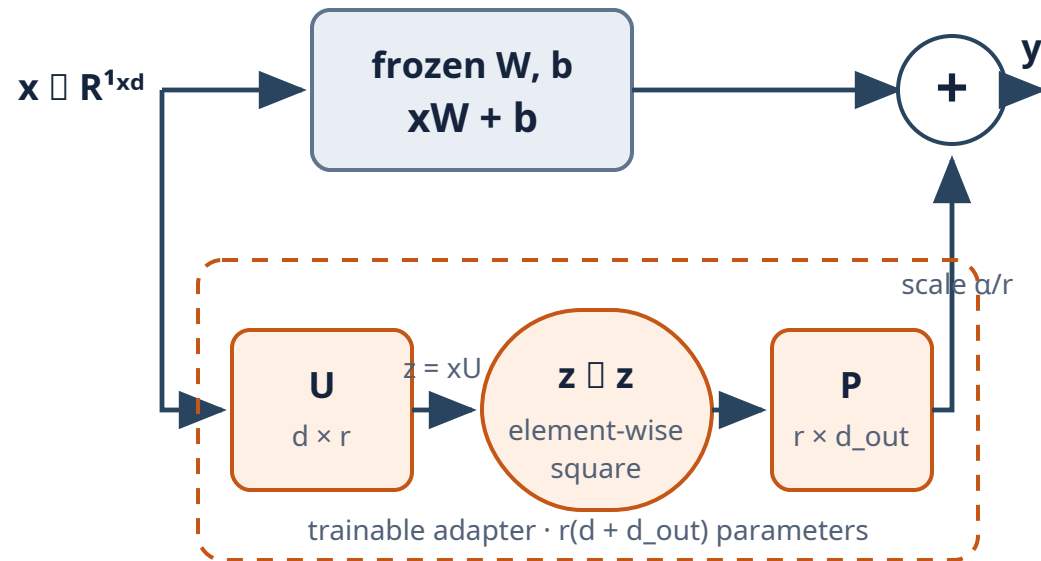
$$\Delta y = (\alpha/r)(xA)B$$



Symmetric Quadratic

activation-dependent quadratic correction

$$\Delta y = (\alpha/r)((xU) \odot (xU))P$$



frozen pretrained path



trainable LoRA path



trainable Symmetric path

For GPT-2 c_proj, $d = d_{out}$, so each adapter has $2dr$ trainable parameters.